

Sumukh Vasisht Shankar

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EDUCATION

Northeastern University

Boston, MA

Master of Science in Data Science – **GPA: 3.92/4.00**

April 2023

Relevant Courses: Data Management and Processing, Supervised and Unsupervised ML, Geometric Deep Learning, Natural Language Processing

National Institute of Engineering

Mysuru, India

Bachelor of Engineering in Information Science – **GPA: 9.05/10.00**

Aug 2021

Relevant Courses: Big Data Analytics, Recommender Systems, Artificial Intelligence, Graph Theory, Neural Networks

RESEARCH EXPERIENCE

Yale University

New Haven, CT

Data Scientist, Cardiovascular Data Science Lab
(M&P Level 26)

Feb 2026 – Present

Data Scientist, Cardiovascular Data Science Lab
(M&P Level 24)

May 2023 – Feb 2026

- Serve as the technical lead for artificial intelligence research and operations at the Cardiovascular Data Science Lab, translating theoretical AI research into deployable, clinically usable systems across multi-site studies.
- Design, develop, and deploy end-to-end machine learning and natural language processing pipelines for cardiovascular research, including deep learning models for ECG image analysis, retrieval-augmented generation systems for clinical knowledge synthesis, and generative models for synthetic data creation.
- Lead the design and implementation of secure, scalable data architectures and cloud-based infrastructures supporting large, multi-center clinical studies, including NIH-funded DETECT-AS, ID-SHD, PRECARDIA, and TRACE-AI, enabling real-time data ingestion, harmonization, and monitoring across institutions.
- Act as the technical lead for TRACE-AI, a multi-institutional clinical validation study evaluating AI-based tools for detecting transthyretin amyloid cardiomyopathy (ATTR-CM) from routine ECGs and echocardiograms, coordinating AI deployment across Mount Sinai, Houston Methodist, University of Washington, UCSF, University of Virginia, Providence Health, Henry Ford Health, UT Southwestern, Duke University, and Oregon Health & Science University.
- Develop and maintain production-grade web and mobile AI platforms used in active clinical studies, directly supporting participant recruitment, longitudinal data collection, cross-site coordination, and AI-driven clinical analysis in real-world healthcare settings.
- Invented and developed CarDS Scan, the first fully on-device AI-ECG mobile application capable of diagnosing ATTR-CM directly from ECG images without cloud connectivity, enabling use in rural, ambulatory, and resource-limited clinical environments.
- Collaborate extensively with international clinicians, researchers, and engineers across Denmark, the United Kingdom (NHS), the Netherlands, Australia, the Czech Republic, Brazil, and India, supporting global AI model validation, deployment, and study execution.
- Provide technical leadership and mentorship to researchers, clinicians, and trainees by defining AI system architectures, overseeing model development and validation, and ensuring regulatory-ready, privacy-preserving implementations suitable for clinical research and translation.

- Function as the primary bridge between clinical investigators and engineering execution, enabling the lab's transition from purely theoretical AI research to real-world clinical deployment and multi-site operational studies.
- Core member of the team that designed and built the CarDS Research Platform, a secure, scalable system for remote consenting, participant enrollment, and onboarding into clinical studies, enabling decentralized recruitment and study execution across geographically distributed healthcare systems.
- PI: Dr. Rohan Khera

Northeastern University

Boston, MA

Research Apprentice, Geometric Deep Learning Lab

Aug 2022 – Sep 2023

- Conducted original research in geometric deep learning, applying equivariant neural network architectures to problems in fluid mechanics and material sciences where physical symmetries are fundamental.
- Analyzed and formalized scale, translation, and rotation symmetries in fluid dynamics and optical systems, and incorporated these symmetries as inductive biases within deep learning models.
- Designed and implemented novel equivariant neural network architectures that integrate physical priors, improving data efficiency, stability, and physical realism compared to conventional deep learning approaches.
- Contributed to peer-reviewed publications and conference presentations arising from this work, including first-author/co-first-author research outputs in applied machine learning and materials science.
- PI: Dr. Robin Walters

Northeastern University

Boston, MA

Research Assistant, Cybersecurity and Privacy Institute

Jan 2022 – Dec 2022

- Conducted original research on Internet measurement and data localization, designing and executing large-scale measurement campaigns to empirically characterize data residency practices across the European Union.
- Designed and implemented end-to-end data collection and analysis pipelines for network measurement studies, enabling reproducible evaluation of cross-border data flows and regulatory compliance behaviors.
- Developed and evaluated machine learning models to infer the physical locations of IP addresses, contributing methodological advances to privacy-preserving Internet measurement research.
- Contributed to peer-reviewed research outputs and manuscripts arising from this work, supporting empirical cybersecurity and privacy studies at the intersection of network measurement and regulation.
- PI: Dr. David Choffness, Advisor: Dr. Alexander Gamero Garrido

Northeastern University

Boston, MA

Research Data Scientist, Lazer Lab

Nov 2021 – May 2023

- Designed, implemented, and maintained scalable data pipelines for large-scale analysis of social media data, enabling real-time characterization of social behavior and information dynamics during major events.
- Developed and validated machine learning models to classify social media content as misinformation or disinformation, supporting empirical research on information integrity and online influence.
- Built interactive misinformation monitoring dashboards that provided real-time analytical insights and visualizations, enabling researchers and stakeholders to track narrative spread and engagement patterns across platforms.
- Worked closely with interdisciplinary research teams to ensure methodological rigor, data quality, and system reliability, improving the robustness and interpretability of large-scale social data analyses.
- PI: Dr. David Lazer

PUBLICATIONS

1. Khunte, A., Sangha, V., Oikonomou, E.K., Dhingra, L.S., Aminorroaya, A., Coppi, A., **Vasisht Shankar, S.**, Rockers, E., Mortazavi, B.J., Bhatt, D.L. and Krumholz, H.M., 2026. Artificial intelligence-based automated interpretation of images of electrocardiograms: development and multinational validation of ECG-GPT. *European Heart Journal-Digital Health*, p.ztag031.
2. Oikonomou, E. K., Craig, N. J., Holste, G., **Vasisht Shankar, S.**, White, A., Mahendran, M., Newby, D. E., Dweck, M. R., & Khera, R. (2025). Artificial intelligence-enabled echocardiography as a surrogate for multimodality aortic stenosis imaging: Post hoc analysis of a clinical trial. *Circulation. Cardiovascular Imaging*, e018353. <https://doi.org/10.1161/circimaging.125.018353>
3. Oikonomou, E.K., Sangha, V., **Vasisht Shankar, S.**, Coppi, A., Krumholz, H.M., Nasir, K., Miller, E.J., Gallegos Kattan, C., Al-Mallah, M.H., Al-Kindi, S. and Khera, R., 2025. Artificial intelligence-enabled electrocardiography and echocardiography to track preclinical progression of transthyretin amyloid cardiomyopathy. *European Heart Journal*, p.ehaf450.
4. **Vasisht Shankar, S.**, Dhingra, L.S., Aminorroaya, A., Adejumo, P., Nadkarni, G.N., Xu, H., Brandt, C., Oikonomou, E.K., Pedroso, A.F. and Khera, R., 2025. Automated transformation of unstructured cardiovascular diagnostic reports into structured datasets using sequentially deployed large language models. *European Heart Journal-Digital Health*, p.ztaf030.
5. Aminorroaya, A., Dhingra, L.S., Pedroso, A.F., **Vasisht Shankar, S.**, Coppi, A., Khunte, A., Foppa, M., Brant, L.C., Barreto, S.M., Ribeiro, A.L.P. and Krumholz, H.M., 2025. Development and multinational validation of an ensemble deep learning algorithm for detecting and predicting structural heart disease using noisy single-lead electrocardiograms. *European Heart Journal-Digital Health*, p.ztaf034.
6. Dhingra, L.S., Aminorroaya, A., Sangha, V., Pedroso, A.F., **Vasisht Shankar, S.**, Coppi, A., Foppa, M., Brant, L.C., Barreto, S.M., Ribeiro, A.L.P. and Krumholz, H.M., 2025. Ensemble deep learning algorithm for structural heart disease screening using electrocardiographic images: PRESENT SHD. *Journal of the American College of Cardiology*, 85(12), pp.1302-1313.
7. **Vasisht Shankar, S.***, Wang, R. *, D'Souza, D., Singer, J.P. and Walters, R. 2024. "Equivariant Neural Networks for Controlling Dynamic Spatial Light Modulators." Integrating Materials and Manufacturing Innovation, November. <https://doi.org/10.1007/s40192-024-00383-1>.
8. Aminorroaya, A., Dhingra, L.S., Oikonomou, E.K., Saadatagah, S., Thangaraj, P., **Vasisht Shankar, S.**, Spatz, E.S. and Khera, R., 2024. Development and multinational validation of an algorithmic strategy for high Lp (a) screening. *Nature Cardiovascular Research*, 3(5), pp.558-566.
9. **Vasisht Shankar, S.**, D'Souza, D., Singer, J.P. and Walters, R., Controlling Dynamic Spatial Light Modulators using Equivariant Neural Networks. In Workshop on "Machine Learning for Materials" ICLR 2023.

PRE-PRINTS

1. Thangaraj, P.M.*, **Vasisht Shankar, S.***, Huang, S., Nadkarni, G., Mortazavi, B., Oikonomou, E.K. and Khera, R., 2024. A Novel Digital Twin Strategy to Examine the Implications of Randomized Control Trials for Real-World Populations. medRxiv. (Accepted for publication – *NPJ Digital Medicine*)
2. Adejumo, P., Thangaraj, P.M., **Vasisht Shankar, S.**, Dhingra, L.S., Aminorroaya, A. and Khera, R., 2024. Retrieval-Augmented Generation for Extracting CHA2DS2VASc Features from Unstructured Clinical Notes in Patients with Atrial Fibrillation. medRxiv, pp.2024-09.
3. Gamero-Garrido, A., Yu, K., **Vasisht Shankar, S.**, Singh, S.K., Balasubramanian, S., Wilcox, A. and Choffnes, D., 2025. Empirically Measuring Data Localization in the EU. *arXiv preprint arXiv:2504.09019*.
4. Aminorroaya, A., Dhingra, L.S., Sangha, V., Oikonomou, E.K., Khunte, A., **Vasisht Shankar, S.**, Camargos, A.P., Haynes, N.A., Hofer, I., Ouyang, D. and Nadkarni, G.N., 2023. Deep learning-enabled detection of aortic stenosis from noisy single lead electrocardiograms. medRxiv, pp.2023-09.

5. **Vasisht Shankar, S.**, Oikonomou, E.K. and Khera, R., 2023. CarDS-Plus ECG Platform: Development and Feasibility Evaluation of a Multiplatform Artificial Intelligence Toolkit for Portable and Wearable Device Electrocardiograms. medRxiv.

*Equal Contribution

PRESENTATIONS

1. CarDS-Scan: On-Device AI for Scalable, Privacy-First ECG Interpretation, ML4Health Symposium, San Diego, California, December 2025; **Spotlight Talk**
2. A Novel Application of Open-Source Large Language Models to Detect Visible and Hidden Clinical Features in 12-Lead Electrocardiograms, ACC Scientific Sessions, Chicago, Illinois, March 2025; **Poster Presentation**
3. Using LLMs to extract tabular EHR data of heart failure cohorts across multicenter, non-interoperable health systems, AHA Scientific Sessions, Chicago, Illinois, November 2024; **Moderated Poster Presentation**
4. Digital Twins for Cardiovascular Care, IEEE BHI 2024, Houston, Texas, November 2024; **Invited Speaker**
5. Generative AI in Healthcare, Yale Healthcare Data Science Workshop, New Haven, Connecticut, September 2024; **Invited Speaker**
6. Equivariant Neural Networks for Controlling Dynamic Spatial Light Modulators. TMS AI in Materials International Conference, Cleveland, Ohio, June 2024; **Oral Presentation**
7. CarDSPlus ECG Platform. AI in Medicine Symposium, Yale University, February 2024; **Poster Presentation**
8. A Novel Digital Twin Strategy to Examine the Implications of Randomized Control Trials for Real-World Populations. ML for Health Conference, New Orleans, Louisiana, December 2023; **Poster Presentation**
9. Controlling Dynamic Spatial Light Modulators using Equivariant Neural Networks. ML4Materials Workshop, The International Conference on Learning Representations (ICLR) (Remote); **Poster Presentation**
10. Controlling Dynamic Spatial Light Modulators using Equivariant Neural Networks. Khoury Research Apprenticeship Showcase, Boston, MA; **Oral Presentation**

PUBLISHED ABSTRACTS

1. **Vasisht Shankar, S.**, Thangaraj, P., Adejumo, P. and Khera, R., 2024. Scalable Phenotyping of Heart Failure Across Multicenter, Non-Interoperable Health Systems Using Retrieval-Augmented Generation and Large Language Models. *Circulation*, 150, pp.A4145446-A4145446.
2. Oikonomou, E.K., Craig, N., Holste, G., **Vasisht Shankar, S.**, White, A., Pawade, T., Newby, D.E., Dweck, M. and Khera, R., 2025. AN AI-ECHO STRATEGY TO DEFINE A PERSONALIZED AORTIC STENOSIS PROGRESSION PROFILE AND ITS PATHOPHYSIOLOGICAL CORRELATES IN THE SALTIRE-2 TRIAL. *Journal of the American College of Cardiology*, 85(12_Supplement), pp.1989-1989.
3. Adejumo, P., Thangaraj, P., Biswas, D., **Vasisht Shankar, S.**, Camargos, A.P. and Khera, R., 2025. A FULLY AUTOMATED ELECTRONIC CLINICAL QUALITY MEASUREMENT FOR HEART FAILURE USING RETRIEVAL AUGMENTED GENERATION AND LARGE LANGUAGE MODELS (LLMS). *Journal of the American College of Cardiology*, 85(12_Supplement), pp.1372-1372.
4. **Vasisht Shankar, S.**, Dhingra, L., Oikonomou, E.K. and Khera, R., 2025. A NOVEL APPLICATION OF OPEN-SOURCE LARGE LANGUAGE MODELS TO DETECT VISIBLE AND HIDDEN CLINICAL FEATURES IN 12-LEAD ELECTROCARDIOGRAMS. *Journal of the American College of Cardiology*, 85(12_Supplement), pp.2845-2845.
5. Adejumo, P., Thangaraj, P., **Vasisht Shankar, S.**, Camargos, A.P., Aminorroaya, A. and Khera, R., 2025. IMPROVING ATRIAL FIBRILLATION RISK ASSESSMENT WITH A MULTIMODAL RAG SYSTEM FOR COMPREHENSIVE CHADS-VASC AND HAS-BLED SCORING. *Journal of the American College of Cardiology*, 85(12_Supplement), pp.2851-2851.

6. Dhingra, L., **Vasisht Shankar, S.**, Biswas, D., Aminorroaya, A., Sangha, V., Camargos, A.P., Oikonomou, E.K. and Khera, R., 2025. AUTOMATED DETECTION OF OCCLUSIVE MYOCARDIAL INFARCTION FROM IMAGES OF ELECTROCARDIOGRAMS USING DEEP LEARNING. Journal of the American College of Cardiology, 85(12_Supplement), pp.1756-1756.
7. Dhingra, L., **Vasisht Shankar, S.**, Aminorroaya, A., Biswas, D., Camargos, A.P., Oikonomou, E.K. and Khera, R., 2025. CATHTAB-LM: AN EFFICIENT LARGE LANGUAGE MODEL FOR AUTOMATED TRANSFORMATION OF UNSTRUCTURED CARDIAC CATHETERIZATION REPORTS INTO STRUCTURED DATASETS. Journal of the American College of Cardiology, 85(12_Supplement), pp.2539-2539.
8. Dhingra, L., Biswas, D., Khunte, A., Sangha, V., **Vasisht Shankar, S.**, Aminorroaya, A. and Khera, R., 2024. Automated Detection of ST-elevation Myocardial Infarction in High-Acuity Settings Using Artificial Intelligence Applied to Images of 12-lead Electrocardiograms. Circulation, 150(Suppl_1), pp.A4145329-A4145329.
9. Adejumo, P., Thangaraj, P., **Vasisht Shankar, S.** and Khera, R., 2024. Retrieval-Augmented Generation for Extracting CHADS-VASc and HAS-BLED Risk Factors from Unstructured Clinical Notes in Patients with Atrial Fibrillation. Circulation, 150(Suppl_1), pp.A4147664-A4147664.
10. Dhingra, L., Khunte, A., **Vasisht Shankar, S.** and Khera, R., 2024. Detecting ST Elevation Myocardial Infarction on a Noisy Single Limb-lead ECG: An Artificial Intelligence-enabled Approach Adaptable to Portable Devices. Circulation, 150(Suppl_1), pp.A4145357-A4145357.
11. Aminorroaya, A., Dhingra, L., Khunte, A., Sangha, V., Pedroso Camargos, A., **Vasisht Shankar, S.**, Sen, S., Haynes, N., Brant, L., Barreto, S. and Ribeiro, A., 2024. Development and Multinational Validation of an Ensemble Deep Learning Algorithm for Detecting Structural Heart Diseases Using Noisy Single-lead Electrocardiograms. Circulation, 150(Suppl_1), pp.A4146465-A4146465.

HONORS & AWARDS

1. Graduate Student Award for Significant Research Impact, Khoury College of CS, 2023
2. Global Nomination and Recognition, NASA SpaceApps Challenge, 2020
3. Mysuru Physics Lecturers Association Award for Excellence in Physics, 2017
4. All India Rank 1 in RIE Physics and Mathematics Stream Exam, 2017

GRANTS & FELLOWSHIPS

1. ALHI Early Career Travel Grant, ML4H Conference, 2023
2. Khoury Research Apprenticeship Fellowship (\$14,000), 2022

TEACHING EXPERIENCE

National Institute of Engineering

Mysuru, India

Teaching Assistant, Dept. of Information Science

Sep 2019 – Dec 2019

- Assisted Prof. CK Vanamala in lectures for Data Analytics and conducted hands on workshops for building recommender systems to a cohort of 75 students.

NIE IEEE Summer of Code

Mysuru, India

Lead Resource Person

Oct 2019

- Resource person for a 1-day workshop for a technical fest - Rubix 2019. Taught the ways of game development using Python and helped the attendees develop a total of six games.

NIE IEEE Student Branch

Mysuru, India

Resource Person, Game Development in Python

May 2019

- Taught Progressive Web Application development and Data Mining to a cohort of over 50 students over a course 7 sessions.

LEADERSHIP & OUTREACH

- | | |
|---|---------------------|
| 1. Co-organizer at Healthcare Data Science Workshop 2024, Yale University | Sep 2024 |
| 2. Career Peer Advisor, Northeastern University | Sep 2023 - May 2024 |
| 3. Co-Organizer, Biblus 2020, National Institute of Engineering | Feb 2020 |
| 4. Co-Organizer, NISB IEEE Women in Engineering 2019 | Mar 2019 |
| 5. Student Activity Committee Coordinator, NISB | May 2019 – May 2020 |
| 6. Student Branch Networking Coordinator, NISB | May 2018 – May 2019 |

PROFESSIONAL EXPERIENCE

Walmart Inc.	Dallas, TX
<i>Data Engineer III Intern</i>	May 2022 – Aug 2022

- Worked on an initiative to generate real-time reports on Walmart+ customer data, focusing on store signups.
- Built near real-time data streaming pipelines using Apache Spark and Apache Kafka to stream petabytes of click data into Google Cloud Storage.
- Developed comprehensive reports and interactive dashboards using Looker, providing actionable insights, and enhancing decision-making processes for the marketing and operations teams.
- Improved data accessibility and reporting efficiency, allowing for timely analysis and response to customer behavior trends.

FICO	Bangalore, India
<i>Software Engineering Intern</i>	Jan 2021 – Jun 2021

- Implemented and integrated a Regular Expression Validation feature for a Mortgage Decision Management Platform tool, enhancing data accuracy and validation efficiency.
- Developed and deployed multiple APIs to facilitate the conversion of data into various formats for optimal storage across different database systems.
- Successfully increased the project's code coverage by over 11\% through comprehensive regression testing, improving the reliability and maintainability of the codebase. This work significantly contributed to the robustness and functionality of the platform.

Antriksh Labs Pvt. Ltd.	Mysuru, India
<i>AI Researcher</i>	Jan 2020 – Dec 2020

- Initiated active methodological research practices to identify optimal visualization techniques for both structured and unstructured data.
- Conducted qualitative research and developed intelligent data pipelines for an AutoML SaaS product.
- Leveraged distributed systems such as Spark and Hadoop to accelerate data processing and statistical analysis, significantly enhancing the efficiency and effectiveness of data workflows.
- This work contributed to advancing the capabilities of the AutoML platform and improving its user experience.

TECHNICAL SKILLS

Languages: Python, R, SQL, C++, HTML, CSS, JavaScript, Swift

Database: MySQL, PostgreSQL, Google Firebase, Mongo DB

Cloud: Google, AWS, Azure

Data Science: Supervised and Unsupervised ML, Deep Learning, Natural Language Processing, Retrieval Augmented Generation (Tensorflow and PyTorch)

Frameworks: Angular, React, Node, Vue

Data Engineering Tools: Spark, Hadoop, Kafka, Zookeeper, Looker